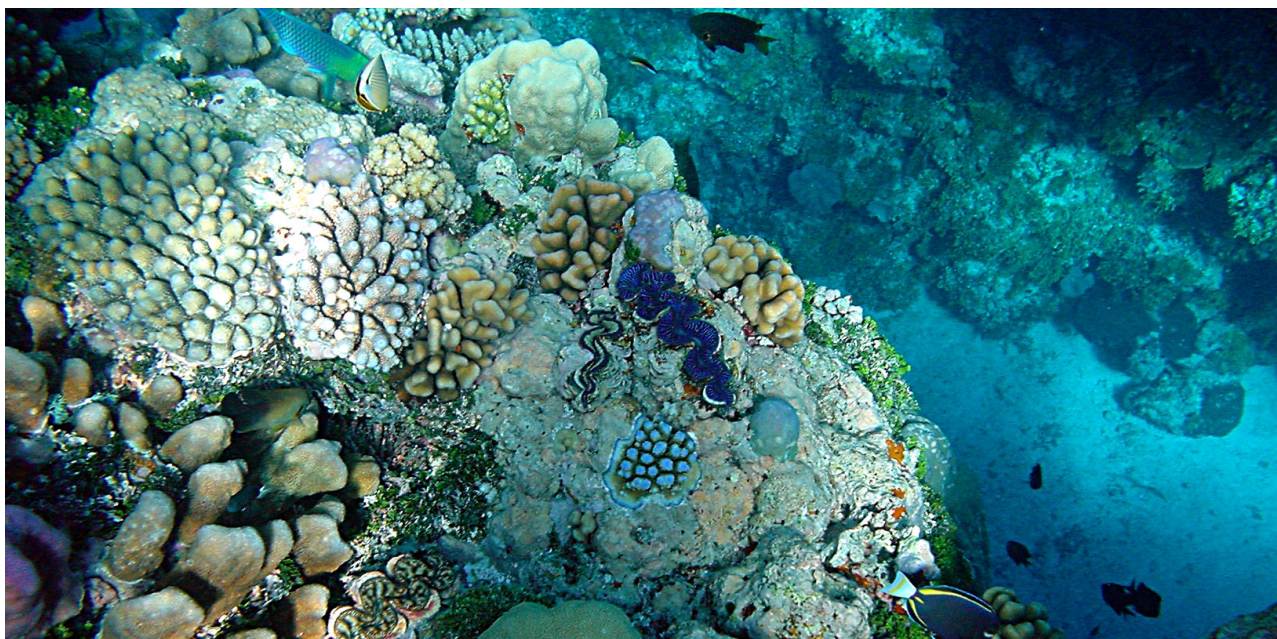


Using Machine Learning to Identify Reefs in the Ocean

Various data and image processing techniques, along with machine learning algorithms, make it possible to study, analyse, understand and identify unknown and lesser-known facts via supervised or unsupervised learning. This article explains the use of PyQGIS to identify reefs in the Indian coastal region.



It was in the middle of the last century that mankind started exploring regions beyond the earth through satellites and space missions. This exploration is broadly classified into *in situ* or onsite observation and remote sensing. The *in situ* observations are limited and the major challenge is to reach out to parts of the Earth and space for monitoring the physical characteristics by acquiring information through reflected and emitted radiation. These can be achieved with the help of sensors and cameras on board satellites, aircraft, unmanned aerial vehicles and drones. Passive remote sensing involves the use of charge coupled devices, radiometers, film photography and infrared. Radars and

LiDARs are active remote sensing elements that emit energy, and scan locations/objects based on the radiation that is reflected or back scattered from the target. The observations are either some data or image.

Geographic information systems (GIS) integrate data, location, maps and information, thereby providing a platform to understand geographical patterns. Popular open source GIS tools are quantum GIS (QGIS), geographic

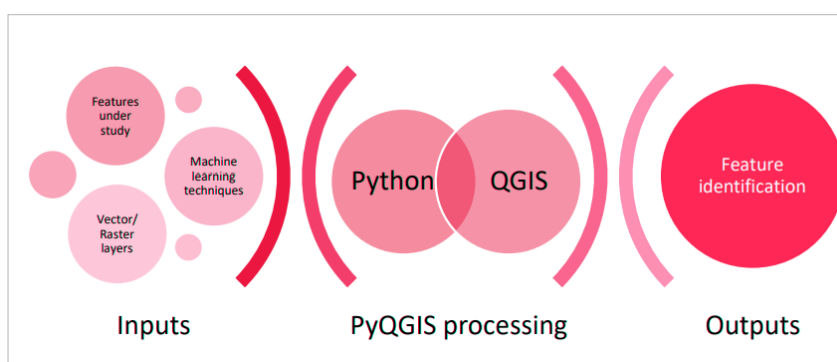


Figure 1: PyQGIS process flow